

If $\mathbf{F}$ is a conservative field, then we know it is a gradient vector field and therefore the Fundamental Theorem for Line Integrals applies. But what happens when $\mathbf{F}$ is *not* conservative? We'd still like analyze phenomena like flow and flux along a simple, closed curve C in the xy -plane:

Of course, we could always parametrize C and compute $\oint_C \mathbf{F} \cdot \mathbf{T} ds$ or $\oint_C \mathbf{F} \cdot \mathbf{n} ds$ directly, but this might not be very efficient. Can we do better? The answer is... yes!

Theorem. [Green's Theorem in the Plane.] If C is a piece-wise smooth, positively oriented¹ contour which encloses a simply connected region R , and $\mathbf{F} = \langle M(x, y), N(x, y) \rangle$ is a vector field on R , then

$$\oint_C \mathbf{F} \cdot \mathbf{T} ds = \oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA.$$

and

$$\oint_C \mathbf{F} \cdot \mathbf{n} ds = \oint_C M dx - N dy = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dA.$$

Key observation: Just like the Fundamental Theorem of Calculus, Green's Theorem says

Give me only information about a function on the *boundary* of a region, and I can tell you function information about *all* of your region.

¹Meaning the direction of travel has the interior on its left. In the plane, this means we travel counterclockwise.

Note that, if $\mathbf{F}$ were conservative, then $\text{curl}(\mathbf{F}) = 0$. One can verify that this implies that $N_x - M_y = 0$ and therefore Green's Theorem says

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA = \iint_R 0 \, dA = 0.$$

Since C is closed, this is exactly what we'd expect from the Fundamental Theorem of Line Integrals.

Example. Use Green's Theorem to find the counterclockwise circulation and outward flux for the field $\mathbf{F}(x, y) = \langle -y^2, xy \rangle$ and C given by the square cut from the first quadrant by the lines $x = 1$ and $y = 1$.

Example. Use Green's Theorem to find the counterclockwise circulation and outward flux for the field $\mathbf{F}(x, y) = \langle y^2 - x^2, x^2 + y^2 \rangle$ and the curve C given by the triangle bounded by $y = 0$, $x = 3$, and $y = x$.

Example. Let C trace the circle $(x - 4)^2 + (y - 2)^2 = 9$ once in the counterclockwise direction. Calculate the line integral

$$\oint_C (7y + x) \, dx + (y + 5x) \, dy, .$$

Example. Consider $P(0,0)$ and $Q(1,1)$. Let C_1 be the path from P to Q along the curve $y = x^2$ and let C_2 be the path from Q to P along the curve $x = y^3$. Now let $C = C_1 \cup C_2$ and calculate

$$\oint_C xy^2 dx + (x^2y + 5x) dy.$$

Of note is that Green's Theorem can be used 'the other way' to compute areas in terms of line integrals. Indeed, if we take any vector field $\mathbf{F} = \langle M(x, y), N(x, y) \rangle$ such that $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 1$, then the circulation form of Green's Theorem reduces to

$$\text{Area of } R = \iint_R 1 dA = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA = \oint_C M dx + N dy.$$

If we choose, for example, $\mathbf{F} = \langle -\frac{y}{2}, \frac{x}{2} \rangle$, then we get

$$\text{Area of } R = \oint_C M dx + N dy = \frac{1}{2} \oint_C x dy - y dx.$$

Example. Consider the region R in the xy -plane bounded by the unit circle C parametrized by $\mathbf{r}(t) = \langle \cos(t), \sin(t) \rangle$ for $0 \leq t \leq 2\pi$. Use Green's Theorem to verify that the area of R is equal to $\pi(1)^2 = \pi$.

Key Observation:

Let $f(x, y)$ be a surface. If $\mathbf{F} = \langle M(x, y), N(x, y) \rangle$ is chosen so that $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = f(x, y)$, then

Green's Theorem says that $\iint_R f(x, y) dA = \oint_C M dx + N dy$. That is, the volume under f over R

can be computed by integrating *only* over C , the boundary of the domain region R . This is *exactly* a higher-dimensional analogue to the single-variable Fundamental Theorem of Calculus!